

The Price of Accountability: Democratic Institutions and the (In)Stability of Economic Data

July 2026

Abstract

Political scientists often judge an economic statistic by its validity (how closely it tracks reality), neglecting its reliability (whether it reports the same value across releases). Yet reliability is political, not just technical. Revising a published statistic like GDP signals that public officials were wrong, inviting accusations of incompetence or manipulation that opposition parties can exploit. No government wants to pay this cost. Using 104 vintages of the World Development Indicators (1994–2021), I show that revisions are more frequent and larger for data collected by democracies, where press and academic freedom empower experts to scrutinize official statistics and demand corrections. This matters for research: the numbers reported for a democracy differ more across releases than those reported for an autocracy.

1 Introduction

In August 2024, the United States Bureau of Labor Statistics (BLS) revised its preliminary employment data, finding that 818,000 fewer jobs had been created in March 2024 than initially reported. This scheduled annual benchmark revision came as no surprise to experts.

Initial job estimates often rely on incomplete household and business surveys; as survey funding and response rates decline, the uncertainty around these estimates tends to increase. In relative terms, the magnitude of this revision was marginal: it meant that 159.2 million individuals, rather than 160 million, were employed in March 2024. However, in the wake of the revision, then-presidential candidate Donald Trump accused then-president Joe Biden of “fraudulently manipulating job statistics,” an opinion echoed by several others in the Republican Party.¹

One year later, in August 2025, the BLS again published a downward revision of its preliminary employment data. That same day, President Trump fired the BLS commissioner, claiming on social media that the revisions had been “RIGGED in order to make the Republicans, and ME, look bad.” Kevin Hassett, the director of the White House National Economic Council, complained that the BLS had been “revising numbers all over the place ... I don’t think anybody really can trust that the numbers are right.”²

The US is not unique in facing controversy over economic statistics. In 2009, Greece revised its planned deficit ratio from 3.7 to 12.5 percent of the Gross Domestic Product (GDP), triggering credit rating downgrades and a loss of market access that brought the country close to default ([Aragão and Linsi, 2022](#)). Between 2010 and 2014, Ghana, Kenya, Nigeria, and other African nations reported GDP increases of up to 60 percent after updating accounting methods and incorporating informal activities ([Jerven and Ebo Duncan, 2012](#)) — a shift associated with less generous lending terms, as countries with per capita incomes above a certain threshold lose access to concessional lending ([Kerner et al., 2017](#)). In 2016, Ireland’s reported GDP growth of 26.3 percent (later traced to Apple’s onshoring of intellectual property) drew ridicule as “leprechaun economics” ([Polyak, 2023](#)). These cases may seem unrelated, but they share a common structure: in each one, a government revised its data to bring official statistics closer to economic reality, and paid a political price for

¹Alicia Wallace. “Trump Routinely Calls Economic Data ‘Fake.’ Here’s Why That’s Dangerous.” *CNN*. 26 January 2025.

²Nick Niedzwiedek and Sam Sutton. “Trump Fires Statistics Chief After Soft Jobs Report.” *Politico*. 1 August 2025.

doing so.

Measuring the economy is an iterative process: preliminary estimates are routinely revised to incorporate new data, improve methods, and correct mistakes (Carson et al., 2004). Macroeconomic indicators like GDP, trade, inflation, and unemployment are pieced together from many different sources, and even the most advanced nations cannot get the “correct” numbers up front. Consequently, new data often contradict previous data. Revisions are a statistical best practice, a natural and necessary component of the data production process. The International Monetary Fund (IMF), the World Bank, AFRISTAT, Eurostat, and other international organizations regularly assist countries seeking to improve their data collection and revision processes. The resulting changes are typically minor technical adjustments.

Data quality has two distinct dimensions: validity, which refers to how accurately data capture the underlying phenomenon being measured, and reliability, which refers to how consistently data provide the same information across repeated measurements and releases (McMann et al., 2022). The previous examples reveal a tension between validity and reliability. Revisions improve validity: they bring official estimates closer to the truth. However, they undermine reliability, because numbers that change over time cannot be trusted to mean the same thing from one release to the next. Governments that prioritize validity will update their data each time better information becomes available, even at the cost of reliability. Those prioritizing reliability may resist correction and sacrifice validity, keeping numbers unchanged to avoid any perception of instability. This trade-off is not just technical: it is political.

As the case of the US shows, even comparatively small, technical revisions can have substantial political and economic consequences. Acknowledging past errors can undermine the government’s reputation, imply a low commitment to transparency, trigger market instability, and erode public trust. The general public often misunderstands the purpose of revisions, and opposition parties capitalize on this misunderstanding. As Ghana, Kenya, and Nigeria show, even large *upward* revisions can be costly, weakening the government’s bar-

gaining position in international financial negotiations. Whether large or small, upward or downward, revisions are both a technicality and a political liability. Under what conditions do the technical pressures to revise outweigh the political costs?

Producing official statistics is a two-stage process. Governments first collect and release preliminary estimates, then decide whether to revise the preliminary estimates (a choice shaped by the quality of the original data). I argue that the context of data *collection* is more important than the context of data *revision*. On average, countries with higher state capacity have better-trained statisticians, regular household surveys, and comprehensive administrative data systems that allow them to collect and report higher-quality data in a timely manner, setting the stage for future revisions. While state capacity provides the technical ability to detect and incorporate new information, governments must also be willing to revise. While no government welcomes the political costs that revisions impose, I argue that revisions are more likely when governments face accountability pressures that make the cost of inaction higher than the cost of correction.

These accountability pressures operate primarily through domestic institutions. Democracies tend to have independent statistical offices that publish their data sources and methodology. This, coupled with academic freedom and free speech, empowers experts to scrutinize official records, identify discrepancies, and demand corrections. Through media coverage, public debate, and academic research, experts compel democratic governments to acknowledge errors and correct official estimates. Democracies do not necessarily *want* to revise their data. Still, journalists, academics, and other experts *pressure* democracies to conduct revisions that are technically necessary but politically costly and would otherwise be overlooked. Opposition parties may try to politicize statistical controversies, but it is the credibility of independent experts — not political competition per se — that drives governments to undertake revisions.

Most of the existing research focuses on data validity (e.g. [Martínez, 2022](#); [Magee and Doces, 2015](#)), asking under what circumstances data capture underlying theoretical concepts

with precision and minimal error. My contribution is to examine the second, less explored dimension: reliability.³ Instead of treating validity and reliability as independent properties, I show that they are in tension: the conditions that produce more valid data are also the conditions that produce less reliable data.

First, I outline the threats to validity that create demand for revisions — *why* countries revise their data. Second, I develop a theory of reliability that explains the timing of revisions — *when* countries revise their data. I find support for my theory in an analysis of GDP data published by the World Development Indicators (WDI) between 1994 and 2021: the democracy finding is robust across three measures of regime type (Polyarchy, Polity 2, and Freedom House) and holds for both the likelihood and magnitude of revisions. I conclude by discussing research and policy implications.

My findings speak to the potential trade-off between validity and reliability, the credibility risks associated with data revisions, and the understudied consequences of revisions to the same data source. While inconsistencies *across data sources* have been widely documented (Linsi et al., 2023; Weikmans and Roberts, 2019; Ram and Ural, 2014; Kerner, 2014; Michaelowa and Michaelowa, 2011), I show that inconsistencies *across releases of the same source* may be even more consequential (Goes, 2023; Johnson et al., 2013). Because democracies revise more, the numbers reported for a democracy differ more across releases than those reported for an autocracy. Data reliability is not a neutral technical property but a political outcome shaped by the same institutional forces that determine whether governments are willing to be held accountable at all.

2 Why Revise?

Countries revise their data because the initial release often has limited validity, failing to capture the “correct” information. This section addresses a more specific question: what

³See Fariss et al. (2022) for one of the few attempts to quantify the reliability of GDP, GDP per capita, and population measurements.

makes preliminary data “incorrect” enough to require revision? I describe three common threats to validity: low statistical capacity, strong political incentives, and inadequate data management.⁴

Regarding statistical capacity, national statistical offices (NSOs) are often underfunded, understaffed, use outdated methods, experience frequent turnover, and rely on outdated information about businesses and households. Many of their employees lack basic statistical competence or familiarity with the System of National Accounts (SNA), a global standardization framework (Mejía Guerra et al., 2023). All of this limits their ability to collect, standardize, and disseminate high-quality data.

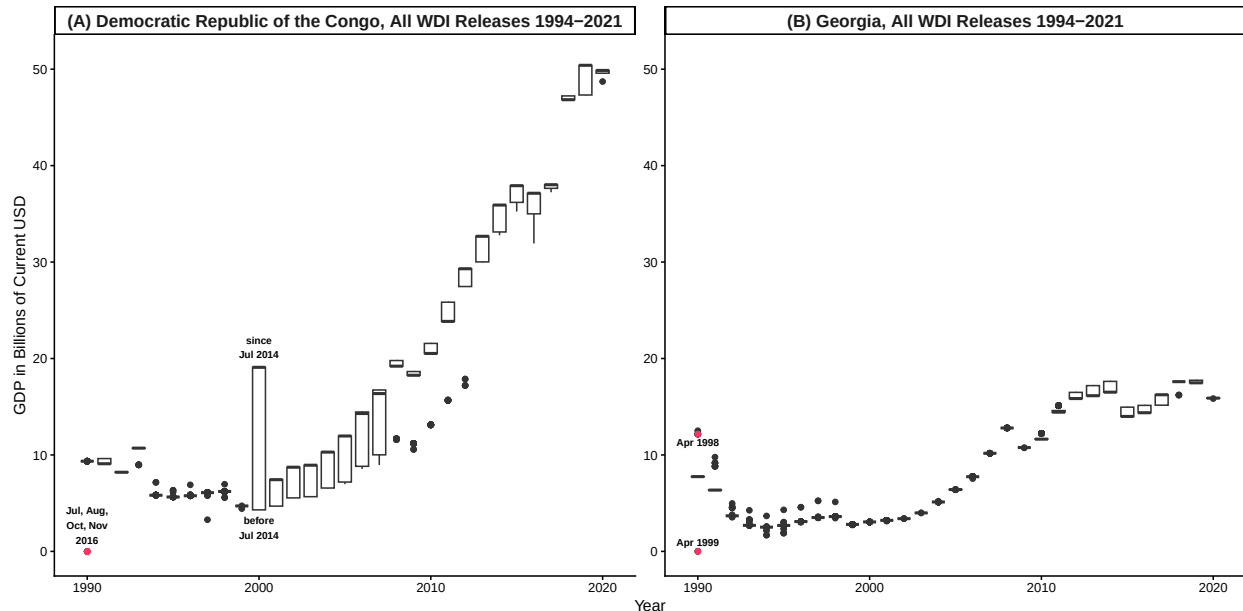
Political incentives also undermine validity. Autocracies are less likely to report policy-relevant data (Hollyer et al., 2011) and may overstate growth rates or underreport COVID-19 deaths (Martínez, 2022; Magee and Doces, 2015; Wallace, 2014; Adiguzel et al., 2020). Political appointees at NSOs might face dismissal if their statistics displease leaders. Returning to this study’s opening example: after the BLS reported unfavorable employment data in August 2025, President Trump dismissed its commissioner and appointed a replacement who proposed ending monthly jobs reports.⁵ Fiscal transfers, aid dependence, climate-conscious electorates, and EU rules create incentives to inflate population figures, understate economic growth, overstate climate aid, and violate deficit limits, respectively (Devarajan, 2013; Kerner et al., 2017; Michaelowa and Michaelowa, 2011; Alt et al., 2014). Conversely, political competition and frequent turnover increase uncertainty about the future, motivating incumbents to pass Freedom of Information (FOI) laws that enhance transparency (Berliner, 2014).

Idiosyncratic errors present a final threat to validity. As the pink dots in Figure 1 show, four WDI vintages report the Democratic Republic of the Congo’s 1990 GDP as *zero*; two others report it as missing. Georgia’s 1990 GDP — reported to be around 12.1707 *billion*

⁴Appendix C describes each validity threat in more detail.

⁵Natalie Sherman. “Trump’s Pick to Lead Economic Data Agency Floats Ending Monthly Jobs Report.” *BBC*. 12 August 2025.

Figure 1: Current GDP of the Democratic Republic of the Congo and Georgia, 1990–2020



These boxplots present the distribution of current GDP estimates from 1990 to 2020 for (A) the Democratic Republic of the Congo and (B) Georgia, using data drawn from the 104 WDI releases from April 1994 to December 2021. Four WDI releases (in pink) reported a GDP of zero for the Democratic Republic of the Congo in 1990. All 32 releases before July 2014 reported a GDP of 4.3 billion for 2000, a figure revised to 19.1 billion in July 2014. Georgia’s GDP in 1990 was reported as 12.1707 *million* in some WDI vintages and 12.1707 *billion* in others (in pink). Section 4.1 describes the data in detail.

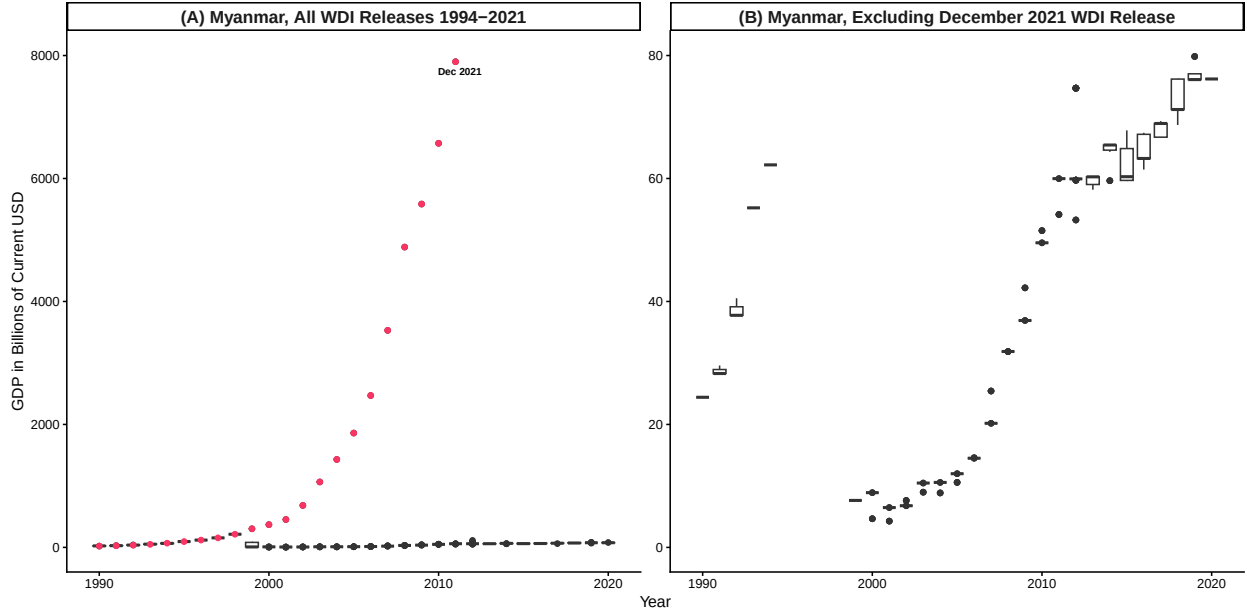
current US dollars until April 1998 — “lost” three digits in the April 1999 and April 2000 vintages, shrinking to 12.1707 *million*.⁶ In the December 2021 update, Myanmar’s 2011 GDP was over 100 times higher than in any other vintage: 7.899 *trillion*, versus 54 to 59 *billion* (Figure 2). Other minor errors include swapped country names, mislabeled columns, and entire indicators mistakenly blanked (see [World Bank 2023](#) for errata). These anomalies are almost certainly the result of idiosyncratic data management errors that are impossible to predict.

Given these threats to validity, international experts advise countries to publish preliminary annual data swiftly, then revise estimates following a regular, publicized schedule that clearly distinguishes between preliminary and updated data.⁷ The reasoning is that

⁶Though Georgia gained formal independence from the Soviet Union in December 1991, its WDI coverage begins in 1990.

⁷As of 2024, 95 percent of IMF members subscribe either to the Special Data Dissemination Standard

Figure 2: Current GDP of Myanmar, 1990–2020



These boxplots present the distribution of current GDP estimates from 1990 to 2020 for Myanmar, using data drawn from the 104 WDI releases from April 1994 to December 2021. The December 2021 WDI release (in pink) is included in (A), but not in (B). As the different y-axes show, the December 2021 release was an outlier, reporting exceptionally high values for the entire time series. Section 4.1 describes the data in detail.

timely macroeconomic data improve governance by enabling more informed policymaking (Islam, 2006). For example, the IMF-produced Report on the Observance of Standards and Codes (ROSC) Data Module⁸ praises countries that revise regularly and reprimands those that do not: “Studies and analyses of revisions should be conducted routinely and used to inform statistical processes and data users” (Sweden, 2001). “Even preliminary data, with the understanding that these are subject to revisions, would be useful” (Uruguay, 2001). “No revision studies are conducted for national accounts and BOP [Balance of Payments] statistics, although they would usefully inform the statistical processes” (Chad 2007). These

(SDDS) or the enhanced General Data Dissemination System (e-GDDS). SDDS subscribers must disseminate preliminary national accounts data with a lag of no more than one quarter after the end of the reference period. Subscribers of the less demanding e-GDDS do not have a hard cutoff. These preliminary estimates should not be confused with flash or advance estimates, which many industrialized nations release within 30 days of the end of each quarter, based on incomplete data.

⁸Most ROSC Data Modules were conducted between 1999 and 2010, and 98 percent of them have been published. Published reports are available at <https://www.imf.org/en/Publications/rosc?sortBy=Topic&sortVal=Data%20Dissemination>

reports are voluntary: countries *requested* an assessment of their data practices, reflecting a widespread interest in increasing statistical capacity.

In these reports, the underlying assumption is that revisions *increase* the validity of preliminary, imperfect data. This assumption could be wrong: revisions could *decrease* validity. However, governments rarely seem to falsify data ex post (Gurieiev and Treisman, 2019). Political leaders might withhold statistics (Venezuela since 2015), postpone the initial release (Zimbabwe in 2019), release doctored numbers from the outset (Argentina in 2007–2008), or erase already-published statistics from public records (the US in 2025), but do not tend to replace “correct” information with falsified information.

In sum, the threats to validity are well-documented, and the international best practices are clear: governments should release preliminary data promptly and revise it regularly as better information becomes available. And yet, as the cases discussed above illustrate, governments frequently resist doing so. This resistance is not arbitrary. Revisions impose real political costs: they signal that previous estimates were wrong, invite accusations of incompetence or manipulation, and hand opposition parties a ready-made narrative about governmental failure. A government that never revises its data avoids these costs entirely. Its numbers may be less valid, but they are consistent — and consistency has political value of its own.

International standards reflect an awareness of this trade-off between validity and reliability. To some extent, ROSC Data Modules consider it tolerable to pursue validity at the expense of reliability: “Based on annual figures, the discrepancy in recent years has generally been within a very acceptable range of less than 1% of GDP” (Estonia, 2001). Yet the same institutions that encourage revisions also recognize their reputational risks, warning that large corrections can signal deep institutional problems, including a lack of “independence, integrity and accountability of the national statistical authorities” (European Commission, 2010, 3).

The central puzzle, then, is not why some governments produce imperfect data. Given

the technical challenges of measuring a modern economy, imperfect data are inevitable. The puzzle is why some governments correct their imperfect data while others do not. If revisions are both statistically necessary and politically costly, something must overcome governments' baseline preference for inertia. The next section develops a theory of when and why that happens, arguing that the answer lies not in the moment of revision itself, but in the institutional conditions under which the original data were collected.

3 How Accountability Pressures Shape Data Revisions

What overcomes governments' baseline preference for inertia? I argue that the answer lies in accountability pressures that raise the political cost of *not* revising above the political cost of revising. Still, accountability pressures interact with a two-stage data production process in which the conditions of the first stage (when data are initially collected and disseminated) shape the feasibility and likelihood of corrections in the second stage (when previously published data may be revised). This sequencing is crucial: strategies adopted at the first stage are difficult to reverse, because they determine the scope of what can be corrected later. A government that collects opaque, poorly documented data forecloses the possibility of meaningful revision later, however strong the pressures it faces. A government that collects transparent, well-documented data makes revisions both technically feasible and externally demanded.

State capacity sets the foundation. Countries with higher state capacity can train and retain skilled personnel, conduct regular surveys, digitize administrative records, audit past estimates, and facilitate cooperation between agencies (for example, between the NSO, tax agency, and finance ministry). All this enhances the quality of initial data, making subsequent revisions easier to implement. Capacity determines whether revision is technically feasible, but not whether governments are willing to bear its political costs. Several high-income nations that could afford to release and revise their data choose not to ([Williams](#),

2009). A report by the [African Development Bank \(2013, 5\)](#) underscores this point: “rich countries are not guaranteed to have good statistics; neither are poor countries condemned to have bad data ... Achieving high-quality data entails a political choice and a firm commitment to invest in statistics.” What determines whether that political choice is made is not capacity itself, but the accountability pressures to which I now turn.

Democracies and constrained executives release more data ([Hollyer et al., 2011](#); [Williams, 2009](#)) and of higher validity ([Magee and Doces, 2015](#); [Martínez, 2022](#)). Leaders facing stiff political competition and regular turnover are more likely to institutionalize transparency through FOI laws, securing future access to government information ([Berliner, 2014](#)). According to [Brambor et al. \(2020\)](#), democracies are better at collecting and processing data not because of political competition, but because of expanded suffrage: states must collect fine-grained information to enable broad popular participation. Democracy is associated not only with more and better data, but also — I argue — with a greater likelihood of data revisions. Beyond the *ability* to revise, countries must be *compelled* to do so. Democracy is the primary institutional mechanism for raising the political cost of inertia and overcoming governments’ reluctance to revise. Democracies enable the work of experts who serve as accountability agents, pressuring for regular revisions no matter how high the political cost.

Democracies are more willing to publicly release raw data, codebooks, and data collection protocols; to retain experienced data managers rather than replace them with political appointees; and to establish external advisory committees that provide technical guidance and independent oversight. This makes their data easier to correct in the second stage. This does not happen due to an intrinsic commitment to transparency. Given the high political cost of data corrections, no government — democratic or authoritarian — would like to admit its mistakes. Still, democracies have little choice: their institutions generate incentives to release transparent data *ex ante* as well as ongoing pressure to revise these data *ex post*.

In terms of the mechanism, I expect revisions to be driven not by political competition but by technocratic expertise. Through media coverage, public debate, and academic

research, experts continuously pressure democratic governments to acknowledge errors and correct official estimates. One possibility is that democracies face such intense transparency pressures that they rush to release incomplete data, creating a greater need for subsequent revisions. Another possibility is that they are more cautious and initially publish understated estimates, as opposed to dictators, who — according to [Martínez \(2022\)](#) — tend to exaggerate their numbers. Either way, democratic leaders have limited control over revisions, which will be as frequent and as large as experts deem necessary.

For the same reason, democracy affects not only the frequency but also the magnitude of revisions. Since democratic institutions empower experts to demand corrections regardless of their size, governments cannot limit revisions to small, face-saving adjustments. When the data requires large corrections, experts will push for them.

In contrast, autocracies are less likely to disclose their data sources and methodology. Without checks and balances, the central government might tamper with the NSO’s work, preventing the collection of important evidence. Autocracies like Azerbaijan, Belarus, Oman, or Tajikistan might recognize the instrumental value of data transparency and request ROSC Data Modules, yet the reputational rewards of international standards rarely outweigh the domestic cost of admitting error. In these countries, data releases are strategic and selective — for example, to attract development aid and foreign investment, gain or maintain access to capital markets, monitor internal challenges, and allocate resources to secure elite support ([Hollyer et al., 2018](#)). “Informational autocrats” gain legitimacy by disseminating selective information about economic successes while concealing information about economic failures ([Guriev and Treisman, 2019](#)). They prioritize control over timeliness, delaying or avoiding publication of data deemed politically unsafe. Information collected by autocrats is a black box: the opaque nature of the data-generation process and the centralized control of information leave little room for subsequent public scrutiny. When freedom of speech is limited, journalists and academics may be hesitant to question official statistics or pressure for corrections. In particular, autocracies with low state capacity face neither the technical conditions

that make revision feasible nor the accountability pressures that make it necessary.

Hypothesis 1: Higher democracy levels in the first stage increase the odds of revisions in the second stage.

Hypothesis 2: Higher democracy levels in the first stage increase the magnitude of revisions in the second stage.

My theory is agnostic about the underlying validity of the revised data. Some countries might have rushed to release low-quality preliminary information that must be corrected *ex post*. Others might only need to fine-tune their data, having released high-quality information from the start. Whatever the validity, I argue that revisions are best explained by contextual factors at the first stage. Conditional on state capacity, domestic accountability pressures create incentives to release transparent information from the outset, thereby enabling subsequent revisions.

4 Operationalizing Data Revisions

4.1 Outcome: Current GDP

As the most widely used source of macroeconomic data in political science ([Goes, 2023](#)), the WDI Database Archives provide 104 electronic releases from 1994 to 2021, each representing a snapshot of the same underlying data at a different point in time.⁹ The differences between snapshots are the revisions I seek to explain.

GDP, the value of all final goods and services produced in a country during a specific period, is “the superstar of indicators” ([Hoekstra, 2019](#), 6) and the most ubiquitous measure of national wealth. In 2020, 205 out of 206 economies surveyed by the IMF compiled and

⁹Though all releases since 1989 are available, the indicator of interest is missing from all releases before 1994, and no release is available for 1996.

published annual GDP statistics (Baer et al., 2022); Eritrea was the lone exception. By comparison, only 109 compiled institutional sector accounts, such as deficit, debt, trade, and foreign direct investment (FDI). Therefore, I focus on revisions to GDP data — specifically, the indicator *GDP in current US dollars* (ID NY.GDP.MKTP.CD), the annual “sum of gross value added by all resident producers in the economy.” The production approach is the most widely used and disseminated method for GDP estimation (Baer et al., 2022, 12). Current GDP enables comparisons across releases, though not across countries or over time, as it does not make PPP or inflation adjustments.¹⁰

There are two stages of data production: first, the reference year, when country c collected and disseminated its data ($y = 1990, \dots, 2020$); second, the WDI vintage, when the same country may have revised its data ($v = \text{April 1994, September 1994}, \dots, \text{December 2021}$). Figure 3 illustrates this structure using Myanmar’s GDP in billions of current US dollars as an example. In this figure, each WDI vintage corresponds to a separate spreadsheet containing all available reference years.¹¹ The goal of my analysis is to explain the differences between spreadsheets.

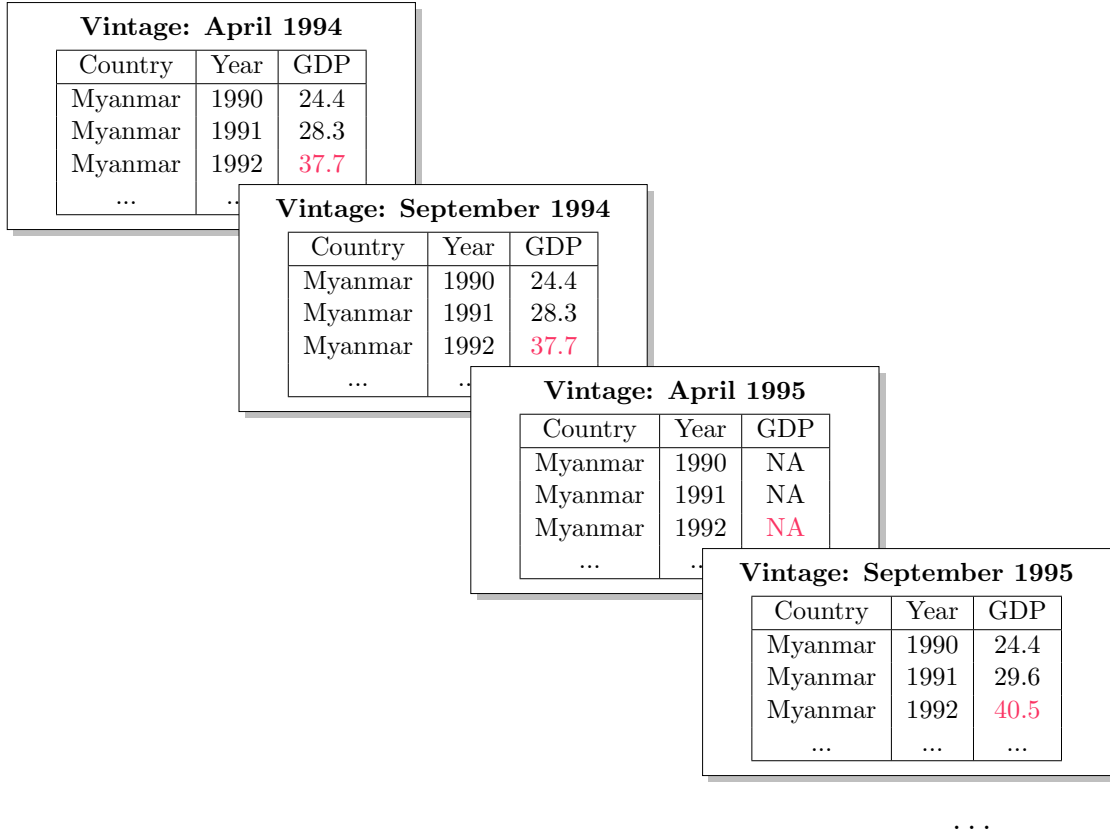
For country c and reference year y , I define a revision as any change in GDP between two consecutive WDI vintages, $v - 1$ and v . Revisions can reflect an addition, deletion, or change of any magnitude, in any direction. As Figure 3 illustrates, Myanmar’s 1992 GDP was reported as 37.7 billion in April 1994, 37.7 billion in September 1994 (*Revision* = 0), missing in April 1995 (*Revision* = 1), and 40.5 billion in September 1995 (*Revision* = 1). Revisions occur in 16.3 percent of all observations.

If a revision occurs and no missing values are involved, I also examine its relative size,

¹⁰ *GDP, PPP (current international \$)* (ID NY.GDP.MKTP.PP.CD) allows for comparisons across countries, not across releases, as the PPP conversion factor changes from one ICP round to another. *GDP in constant US dollars* (ID NY.GDP.MKTP.KD), calculated using the GDP deflator (the ratio of GDP in current local currency to GDP in constant local currency) to account for inflation, allows for comparisons over time, not across releases.

¹¹ To ensure that each reference year has had time to be compiled and published, the analysis includes only vintages released at least 90 days after the end of the reference period. For example, for the reference year 2020, this corresponds to all WDI releases starting in April 2021. I use 90 days as a cutoff because 39.8 percent of all economies surveyed by the IMF in 2020 reported annual GDP data within 90 days of the reference period; 54.4 percent did so within 91 to 365 days (Baer et al., 2022, 17).

Figure 3: Illustration of the Data Structure, Using the Case of Myanmar



This figure uses the case of Myanmar to illustrate the three-dimensional structure of the data: the unit of analysis is country c , year y , and vintage v . In this notation, $y = 1990, \dots, 2020$ refers to the first stage, when the data are initially collected and disseminated, whereas $v = \text{April 1994, September 1994}, \dots, \text{December 2021}$ refers to the second stage (i.e., each subsequent WDI release), when previously published data may be revised.

ranging from a 100 percent reduction (the Democratic Republic of Congo, Figure 1) to a 17,488 percent increase (Myanmar, Figure 2). The average revision is relatively small: 4.2 percent. Since the focus is on magnitude rather than direction, I use the absolute value of percentage changes, logged to reduce the influence of extreme values.

4.2 Empirical Strategy

I estimate two sets of models by maximum likelihood. First, I use logistic regressions to capture the probability that a given country-year observation (say, Myanmar-1992) is revised

between two consecutive WDI vintages:

$$\Pr(\text{Revision}_{c,y,v} = 1) = \text{logit}^{-1}(\alpha + \mathbf{X}_{c,y,v}\beta + u_c + u_y + u_v) \quad (1)$$

where $\text{Revision}_{c,y,v}$ equals one if the GDP value for country c in reference year y differs between vintage $v - 1$ and vintage v ; $\mathbf{X}_{c,y,v}$ denotes a set of independent variables (described in the next section); and u_c , u_y , and u_v are random intercepts for country, year, and vintage, respectively.

Second, conditional on a revision occurring, with no missing values involved, I use linear regressions to estimate its magnitude, measured as the logged absolute percentage change:

$$\text{Absolute \% Change (Log)}_{c,y,v} = \alpha + \mathbf{X}_{c,y,v}\beta + u_c + u_y + u_v + \varepsilon_{c,y,v}, \quad (2)$$

where u_c , u_y , and u_v are random intercepts for country, year, and vintage, respectively, and $\varepsilon_{c,y,v}$ is an idiosyncratic error term. This two-step approach allows me to separately model whether a revision occurs and, if so, its size.¹²

4.3 Independent Variables

My theory focuses primarily on regime type as the main predictor of data revisions, with state capacity serving as a background condition. I measure regime type using three indices. My primary measure, V-Dem’s *Polyarchy* index (Coppedge et al., 2023), captures the quality of electoral democracy (including extensive suffrage, fair elections, freedom of expression, and access to information) on an ordinal scale from 0 to 1. As robustness checks, *Polity 2* (Marshall and Gurr, 2020) ranges from -10 (hereditary monarchy) to $+10$ (consolidated democracy); *Freedom House* originally ranges from 1 (most free) to 7 (least free), a scale I flip so it is consistent with the other two measures. Across all three indices, higher values

¹²I model the two steps separately because Tobit or Heckman selection models assume independent observations and cannot incorporate random intercepts.

indicate more democratic regimes (and thus more domestic accountability pressures).

Why are democracies more subject to public scrutiny? The key mechanisms I propose are *Media Freedom* and *Freedom of Academic Exchange*. The former captures the extent to which governments censor political information in the media; the latter captures the degree to which scholars are free to exchange and communicate research findings, including through media engagement and public lectures. Both are V-Dem indicators (Coppedge et al., 2023), from 0 to 4, with higher values indicating rare censorship and full academic freedom, respectively.¹³ In the absence of censorship and intimidation, experts can scrutinize official statistics and demand transparent data practices. I also assess alternative mechanisms from Berliner (2014): *Opposition Strength* measures the vote share of the largest opposition party in the most recent legislative election, whereas *Turnover Frequency* indicates the number of changes in executive party control over the preceding five years (both from the Database of Political Institutions, Cruz et al. 2021). Following Brambor et al. (2020), who show that suffrage expansion drives investments in information capacity, I include the population share with *Suffrage* as an alternative pathway through which democracy might affect revision behavior through broad popular participation. Finally, I include the *Political Corruption Index* from V-Dem as an alternative mechanism: more corrupt environments may suppress the magnitude of revisions, as officials have incentives to minimize the visibility of corrections rather than acknowledge past errors.

My main proxy for state capacity is *Public Administration*, a V-Dem measure ranging from 0 (the law is not respected by public officials) to 4 (the law is generally fully respected by public officials).¹⁴ This measure is imperfect: it primarily captures bureaucratic impartiality and rule-following, rather than statistical expertise. Still, it has broader temporal and spatial coverage than alternative capacity measures, such as the Hanson and Sigman (2021) state capacity index, the Brambor et al. (2020) information capacity index, and the International

¹³The official variable names in the V-Dem codebook are *Government Censorship Effort — Media* and *Freedom of Academic Exchange and Dissemination*.

¹⁴The official variable name in the V-Dem codebook is *Rigorous and Impartial Public Administration*.

Country Risk Guide measure of bureaucratic quality.¹⁵

Models also include a dichotomous indicator of *IMF Program* participation (Kentikelenis and Stubbs, 2023) as a control for foreign accountability pressure, along with indicators for a *Financial Crisis* (Nguyen et al., 2022), *Natural Disaster* (Centre for Research on the Epidemiology of Disasters, 2020), and *Armed Conflict* (Gleditsch et al., 2002), all of which may independently prompt data revisions.

Three final variables capture technical or methodological factors. *SNA Change* takes the value of one in years when countries adopted or updated the SNA in use, using data from the UN National Accounts Statistics (complemented by WDI Metadata and IMF International Financial Statistics). *Data Management Error* takes the value of one for reporting mistakes documented in the World Bank’s data updates and errata (World Bank, 2023), like those in Figures 1 or 2 (see Appendix C). Lastly, *Log Difference* measures the gap (in years) between the first and second stages, $v - y$, logged to capture the nonlinear decay in revision frequency over time. Revisions are most common shortly after the initial data release and become progressively less frequent as more time elapses (Croushore and Stark, 2003).

Table 1 summarizes all independent variables and their underlying concepts. All substantive predictors are assigned to the first stage (y), consistent with my core argument that the conditions under which data are initially collected matter more than the conditions at the time of potential revision.¹⁶ Vintage-specific variables (SNA changes, data management errors, and time between vintage and year) are assigned to the second stage (v), since they capture technical features of the revision process itself, rather than the institutional environment of data collection. All time-varying predictors are lagged by one year to avoid simultaneity bias.¹⁷

¹⁵As Appendix D shows, the democracy result is robust to replacing *Public Administration* with these alternative measures.

¹⁶Including the same predictors at both stages would introduce severe multicollinearity due to high correlation (for example, *Polyarchy* values at y and v are correlated at $r = 0.88$, $p < 0.001$), making the two sets of coefficients difficult to interpret independently and inflating the model without theoretical justification.

¹⁷I do not include economic indicators like foreign aid, natural resource rents, or trade dependence because they are revised in parallel with GDP.

Table 1: Summary of Independent Variables

Variable	Underlying Concept	Period
Polyarchy	Domestic Accountability	y
Polity 2	Domestic Accountability	y
Freedom House	Domestic Accountability	y
<i>Domestic Accountability Mechanisms:</i>		
– <i>Media Freedom</i>	Mechanism: Media Freedom	y
– <i>Freedom of Academic Exchange</i>	Mechanism: Academic Freedom	y
– <i>Political Corruption Index</i>	Alternative Mechanism	y
– <i>Suffrage</i>	Alternative Mechanism	y
– <i>Opposition Strength</i>	Alternative Mechanism	y
– <i>Turnover Frequency</i>	Alternative Mechanism	y
Public Administration	State Capacity	y
IMF Program	Foreign Accountability	y
Financial Crisis	Exceptional Events	y
Natural Disaster	Exceptional Events	y
Armed Conflict	Exceptional Events	y
SNA Change	Methodological Changes	v
Data Management Error	Idiosyncratic Errors	v
Log Difference	Time Since Initial Data Dissemination	$v - y$

5 Determinants of Data Revisions

5.1 Main Results

Table 2 presents the results of six models. Models 1 to 3 are logistic regressions with the dichotomous outcome *Revision*. Models 4 to 6 are three linear regressions with the continuous outcome *Absolute % Change (Log)*, restricting the sample to instances when *Revision* equals one (hence the smaller number of observations). Each pair of models uses a different measure of regime type: *Polyarchy* (Models 1 and 4), *Polity 2* (Models 2 and 5), and *Freedom House* (Models 3 and 6).

Consistent with Hypothesis 1, higher levels of democracy in the first stage are associated with a higher likelihood of revisions in the second stage. This effect is positive and significant ($p < 0.01$) for all three democracy measures: *Polyarchy*, *Polity 2*, and *Freedom House*. The association is incremental: holding other covariates at their means, moving from 0.25 to 0.75 on the *Polyarchy* scale (roughly the gap between Russia and South Africa in 2020) is associated with about a one-third increase in the likelihood of revision. Higher democracy levels are also associated with larger revision magnitudes, though this result is weaker for

Table 2: Determinants of the Likelihood and Magnitude of Data Revisions

	Dependent Variable:					
	Revision (1)	Revision (2)	Revision (3)	Abs. % Change (4)	Abs. % Change (5)	Abs. % Change (6)
Polyarchy, y	0.56*** (0.09)			1.31*** (0.34)		
Polity 2, y		0.02*** (0.00)			0.02* (0.01)	
Freedom House, y			0.04*** (0.01)			0.09* (0.05)
Public Administration, y	0.03* (0.02)	0.05*** (0.02)	0.07*** (0.02)	0.12 (0.08)	0.19*** (0.07)	0.18** (0.07)
IMF Program, y	0.02 (0.02)	0.02 (0.02)	0.04** (0.02)	-0.13* (0.08)	-0.10 (0.08)	-0.09 (0.08)
Financial Crisis, y	0.00 (0.02)	-0.00 (0.02)	-0.00 (0.02)	-0.17** (0.08)	-0.18** (0.08)	-0.15* (0.08)
Natural Disaster, y	0.05*** (0.02)	0.06*** (0.02)	0.06*** (0.02)	-0.12 (0.07)	-0.13* (0.07)	-0.09 (0.07)
Armed Conflict, y	0.05 (0.03)	0.04 (0.03)	0.05 (0.03)	-0.24* (0.13)	-0.22* (0.13)	-0.24* (0.13)
SNA Change, v	0.23*** (0.03)	0.25*** (0.03)	0.23*** (0.03)	1.70*** (0.12)	1.77*** (0.12)	1.70*** (0.12)
Data Management Error, v	6.28*** (1.16)	6.36*** (1.16)	6.28*** (1.16)	11.52*** (1.49)	11.86*** (1.51)	11.30*** (1.53)
Log Difference, $v - y$	-1.02*** (0.03)	-1.03*** (0.03)	-1.00*** (0.03)	-4.66*** (0.10)	-4.71*** (0.11)	-4.65*** (0.10)
Intercept	-2.48*** (0.56)	-2.25*** (0.56)	-2.18*** (0.56)	-0.30 (0.92)	0.41 (0.90)	0.50 (0.93)
Log Likelihood	-74192.07	-70242.86	-73777.09	-209846.50	-199096.55	-208615.15
Observations	375052	356324	373139	62973	59755	62601
Number of Countries	169	163	169	169	163	169
Number of Years	31	30	31	31	30	31
Number of Vintages	103	103	103	78	78	78
Variance: Countries (Intercept)	0.23	0.24	0.25	3.14	3.43	3.36
Variance: Years (Intercept)	0.04	0.05	0.03	12.64	11.10	12.37
Variance: Vintages (Intercept)	27.92	27.54	27.97	26.13	26.35	25.98

Notes: This table presents three logistic regressions (Models 1–3) and three linear regressions (Models 4–6) with random intercepts for country, year, and vintage. y is the reference year; v is the vintage. Standard errors in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

Polity 2 ($p = 0.052$) and *Freedom House* ($p = 0.056$). The magnitude finding is strongest for *Polyarchy* ($p < 0.01$): the same shift from 0.25 to 0.75 is associated with revisions roughly twice as large. The broader conception of electoral democracy captured by this index (including freedom of expression and access to information) is more closely linked to the size of revisions than the narrower institutional constraints captured by the other two

indicators. The results support Hypothesis 2, though with varying strength across measures.

Among the controls, *Public Administration* has a positive effect on the likelihood of revision, consistent with the expectation that higher bureaucratic capacity facilitates the detection and correction of errors. IMF programs, financial crises, natural disasters, and armed conflicts are generally associated with an increase in the likelihood of revision and a decrease in the magnitude of revision, reflecting disruptions to regular data collection during exceptional events. At time v , SNA changes and data management errors are consistently associated with a higher likelihood of revisions and larger revision magnitudes, as expected. The log difference between vintage and year is negative and highly significant in all six models, confirming that revisions are most common shortly after the initial data release and become progressively less frequent as more time elapses (Croushore and Stark, 2003). Finally, the vintage random intercept absorbs much of the latent variation, indicating that some releases revise many countries at once. This is consistent with releases silently correcting errors that were not officially documented.

5.2 Decomposing the Mechanism

Table 3 decomposes the democracy finding by examining which specific aspects of democratic governance are associated with data revisions. *Media Freedom* is positive ($p < 0.01$) for both outcomes: where governments refrain from directly or indirectly attempting to censor print or broadcast media, revisions are both more frequent and larger. *Freedom of Academic Exchange* has a positive and significant effect on likelihood ($p < 0.01$), but not on magnitude. This asymmetry suggests that academic freedom enables the *decision* to revise, but it does not independently determine *how large* the revision is once the decision has been made.

The association between democracy and revision behavior runs through press and academic freedom. As Model 1 shows, no alternative mechanism is significantly associated with the likelihood of revision. Still, the electoral competition and popular participation mechanisms influence the magnitude, as Model 2 shows. The share of population with *Suffrage*

Table 3: Determinants of the Likelihood and Magnitude of Data Revisions: Which Aspects of Democracy Matter?

	Dependent Variable:	
	Revision (1)	Abs. % Change (2)
Media Freedom, y	0.05*** (0.01)	0.29*** (0.06)
Freedom of Academic Exchange, y	0.09*** (0.02)	-0.16 (0.10)
Suffrage, y	0.09 (0.11)	0.88** (0.45)
Opposition Strength, y	-0.02 (0.05)	-0.56*** (0.20)
Turnover Frequency, Prev. 5 Yrs, y	0.00 (0.01)	0.05 (0.06)
Political Corruption Index, y	0.04 (0.10)	-1.54*** (0.39)
Public Administration, y	0.02 (0.02)	0.06 (0.09)
IMF Program, y	0.02 (0.02)	-0.09 (0.08)
Financial Crisis, y	-0.01 (0.02)	-0.17** (0.08)
Natural Disaster, y	0.06*** (0.02)	-0.11 (0.07)
Armed Conflict, y	0.05 (0.03)	-0.16 (0.14)
SNA Change, v	0.24*** (0.03)	1.81*** (0.12)
Data Management Error, v		11.14*** (1.53)
Log Difference, $v - y$	-0.98*** (0.04)	-4.70*** (0.11)
Intercept	-2.72*** (0.58)	0.18 (1.06)
Log Likelihood	-69537.97	-195506.15
Observations	350360	58655
Number of Countries	166	166
Number of Years	30	30
Number of Vintages	103	78
Variance: Countries (Intercept)	0.24	2.96
Variance: Years (Intercept)	0.02	12.87
Variance: Vintages (Intercept)	28.48	26.63

Notes: This table presents a logistic regression (Model 1) and a linear regression (Model 2) with random intercepts for country, year, and vintage. *Data Management Error* is excluded from Model 1 due to near-perfect separation. y is the reference year; v is the vintage. Standard errors in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

is positive and significant ($p < 0.01$), whereas the vote share of the largest opposition party in the most recent legislative election (*Opposition Strength*) is negative ($p < 0.01$). As the coefficient for *Political Corruption Index* indicates, more corrupt environments significantly suppress the size of revisions even when revisions do occur ($p < 0.01$), consistent with officials having incentives to minimize the visibility of corrections rather than acknowledge past errors fully. The remaining results are similar to those in Table 2.¹⁸ *Public Administration* loses significance, likely because the democracy component indicators capture overlapping aspects of institutional quality that absorb the independent contribution of bureaucratic capacity.

Revisions are most likely when journalists, academics, and other experts have the freedom to demand data transparency and push for data corrections. This finding speaks directly to the validity-reliability trade-off: the institutional conditions that produce more valid data (freedom of academic expression, independent expertise, and the resulting transparency pressures) also produce less reliable data. In contrast, pressure from voters and political rivals is insufficient to increase the likelihood of revision (though these factors do matter for the magnitude of revision).

Additional tests confirm that the democracy finding is not an artifact of measurement choices, sample composition, or model specification. Appendix D presents models with alternative measures of state capacity. The democracy finding remains robust throughout. Appendix E restricts the analysis to the main scheduled annual release (typically in April), reducing noise from the increased update frequency in recent years. The democracy finding holds for likelihood but is weaker for magnitude in this reduced sample, likely due to the substantial reduction in observations. Appendix F replaces random effects with country, year, and vintage fixed effects; the Polyarchy finding survives for likelihood but is less consistent for

¹⁸One difference is that *Data Management Error* is excluded from Model 1 of Table 3 due to separation issues. Due to missing values in the democracy components, the models in Table 3 rely on smaller samples than the equivalent models in Table 2. Idiosyncratic errors are rare events: in a reduced sample, there are too few such observations to reliably identify the coefficient for *Data Management Error* in the logistic regression.

magnitude, which is expected given the well-known efficiency costs of three-way fixed effects models with many parameters. Finally, Appendix G examines the direction of revisions: compared to downward revisions, upward revisions are significantly more likely at higher Polyarchy scores, suggesting that initial reporting in democracies tends to be cautious and that subsequent updates adjust figures upward as more information becomes available.

6 Discussion and Conclusion

This study shows that, conditional on state capacity, revisions are more likely and larger for GDP data collected by democracies. The finding is robust across three alternative measures of regime type: *Polyarchy*, *Polity 2*, and *Freedom House*. Beyond GDP data, I expect these predictions to hold for other macroeconomic indicators, like FDI, trade, inflation, and unemployment, all of which are regularly revised. Beyond WDI data, the argument likely travels to other data sources. Since the Penn World Table (PWT) and the Maddison Project Database are not updated as frequently, they do not allow for as systematic a test.¹⁹ Still, future research can validate the generalizability of my results by exploring variations in the frequency and magnitude of revisions across other indicators and data sources.

These results have four key implications. First, they add nuance to existing evidence that democracies produce data with higher validity. Democratically elected leaders might be less likely to lie about growth rates or COVID-19 deaths (Martínez, 2022; Adiguzel et al., 2020), yet they face greater incentives to scrutinize and revise information. As a result, democracies produce data with lower reliability: their numbers are more likely to change over time. Even revisions undertaken in good faith can create a perception of mismanagement, an issue that is politically costlier for democracies (which are held to higher transparency standards than autocracies). In contrast, autocrats might collect biased data and publish imprecise estimates, but once released, these estimates are rarely revised. For different reasons, all

¹⁹For example, between 1994 and 2024, there have been eleven PWT releases and five releases of the Maddison Project Database, compared to over 100 WDI vintages available in the WDI Database Archives.

data are flawed.

Second, transparency in revision processes must be accompanied by public communication strategies that frame revisions as a sign of statistical rigor. In the US, the backlash against BLS employment data updates in 2024 and 2025 illustrates how, in contexts of low data literacy and widespread misinformation, revisions can erode confidence in official statistics, undermining the government’s authority as the main producer of macroeconomic data and driving individuals toward private data providers.²⁰ Breaking this cycle requires a broader public understanding of why reliable data and valid data are different, and why pursuing one can come at the expense of the other.

Third, the environment of initial data collection matters more than subsequent adjustments. No amount of posterior revision can compensate for flaws in the original data collection process, and present-day improvements to statistical capacity do not automatically fix past data. The sequencing of data quality investments matters as much as their magnitude.

Finally, my results extend work on the discrepancies between WDI and PWT vintages, or between different vintages of the same data source (Goes, 2023; Johnson et al., 2013; Croushore and Stark, 2003). While much of the literature has focused on non-democracies (Goes, 2023) or African states (Johnson et al., 2013), my results suggest that industrial democracies also warrant scrutiny: their data may be more valid but less stable than assumed. When using data from democracies, researchers implicitly trade reliability for validity, benefiting from accountability pressures that produce more accurate estimates while neglecting the instability these pressures introduce. When using data from autocracies, researchers face the opposite trade-off: greater consistency across vintages but at the cost of accuracy. Neither choice is neutral, and researchers should be explicit about which dimension of data quality their analysis depends on most. As McMann et al. (2022) show, discussions about data quality often focus on validity and neglect reliability. This study demonstrates that the two are bound together by the politics of data production — and that understanding

²⁰Geoffrey Morgan and Matthew Griffin. “Wall Street Leaning Harder on Private Data After Trump BLS Spat.” *Bloomberg*. 23 August 2025.

one requires understanding the other.

Declaration of Competing Interests

The author declares no competing interests.

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Appendix for
The Price of Accountability: Democratic Institutions and the
(In)Stability of Economic Data

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A Countries Included in the Main Analysis

Afghanistan, Albania, Algeria, Angola, Argentina, Armenia, Australia, Austria, Azerbaijan, Bahamas, Bahrain, Bangladesh, Barbados, Belarus, Belgium, Benin, Bhutan, Bolivia, Bosnia and Herzegovina, Botswana, Brazil, Bulgaria, Burkina Faso, Burundi, Cabo Verde, Cambodia, Cameroon, Canada, Central African Republic, Chad, Chile, China, Colombia, Comoros, Congo, Costa Rica, Cote d'Ivoire, Croatia, Cuba, Cyprus, Czech Republic, Democratic Republic of the Congo, Denmark, Djibouti, Dominican Republic, Ecuador, Egypt, El Salvador, Equatorial Guinea, Eritrea, Estonia, Eswatini, Ethiopia, Fiji, Finland, France, Gabon, Gambia, Georgia, Germany, Ghana, Greece, Grenada, Guatemala, Guinea, Guinea-Bissau, Guyana, Haiti, Honduras, Hungary, Iceland, India, Indonesia, Iran, Iraq, Ireland, Israel, Italy, Jamaica, Japan, Jordan, Kazakhstan, Kenya, Kiribati, Kuwait, Kyrgyzstan, Laos, Latvia, Lebanon, Lesotho, Liberia, Libya, Lithuania, Luxembourg, Madagascar, Malawi, Malaysia, Maldives, Mali, Malta, Mauritania, Mauritius, Mexico, Moldova, Mongolia, Morocco, Mozambique, Myanmar, Namibia, Nepal, Netherlands, New Zealand, Nicaragua, Niger, Nigeria, North Macedonia, Norway, Oman, Pakistan, Panama, Papua New Guinea, Paraguay, Peru, Philippines, Poland, Portugal, Qatar, Romania, Russia, Rwanda, Saint Vincent and the Grenadines, Samoa, Saudi Arabia, Senegal, Sierra Leone, Singapore, Slovakia, Slovenia, Solomon Islands, Somalia, South Africa, South Korea, South Sudan, Spain, Sri Lanka, Sudan, Suriname, Sweden, Switzerland, Syria, Tajikistan, Tanzania, Thailand, Timor-Leste, Togo, Tonga, Trinidad and Tobago, Tunisia, Turkey, Turkmenistan, Uganda, Ukraine, United Arab Emirates, United Kingdom, United States, Uruguay, Uzbekistan, Vanuatu, Venezuela, Vietnam, Yemen, Zambia, Zimbabwe.

B Variables Included in the Main Analysis

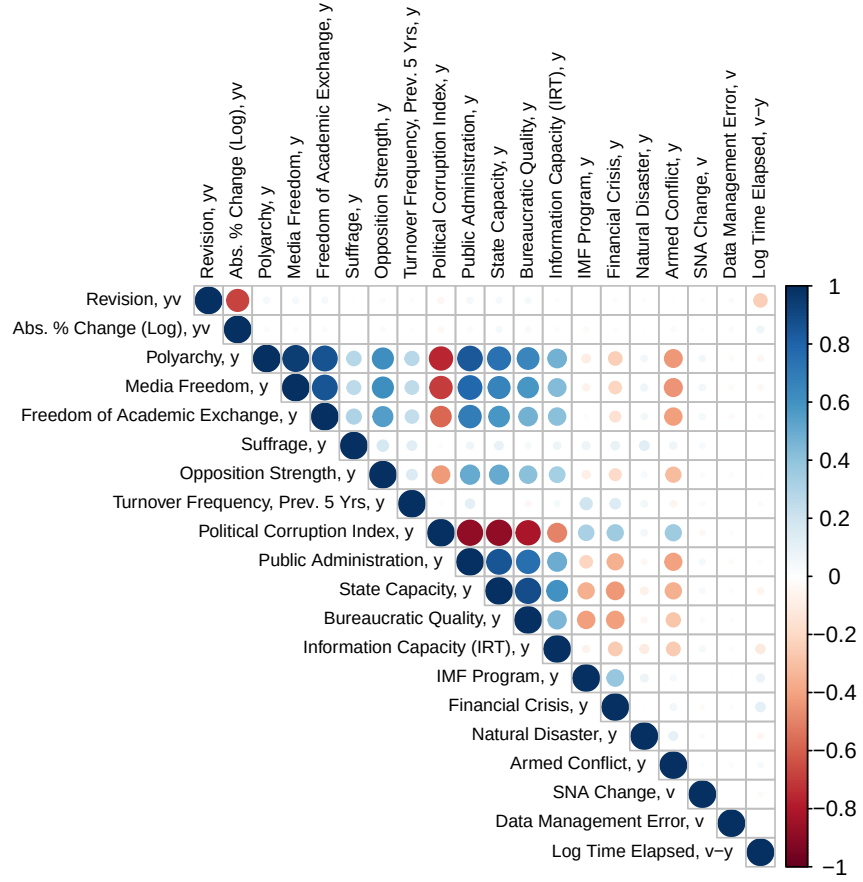
Table B.1: Variables Included in the Main Analysis

Variable	Description	Coverage	Source
Absolute % Change (Log)	Absolute value of the percentage change in GDP between two consecutive WDI releases for the same country-year pair, logged to reduce the influence of extreme values. Defined only when a revision occurs and neither value is missing	1990–2021	WDI
Armed Conflict	Was there an armed conflict? Yes = 1	1990–2021	Gleditsch et al. (2002)
Bureaucratic Quality	To what extent does the country's bureaucracy have the strength and expertise to govern without drastic changes in policy or interruptions in government services? Low = 0, High = 4	1990–2021	The PRS Group (2022)

Data Management Error	Coded 1 for the following observations: Armenia, 1992, April 1998 WDI; China, 2007 and 2008, May 2009 WDI; Democratic Republic of the Congo, 1990, July 2016 to April 2017 WDI; Estonia, 1995, April 1997 WDI; Georgia, 1990, April 1999 and April 2000 WDI; Myanmar, all years, December 2021 WDI	1990–2021	Own coding, based on World Bank (2023)
Financial Crisis	Was there a banking, currency, or debt crisis? Yes = 1	1990–2019	Nguyen et al. (2022)
Freedom House	Combined measure of political rights and civil liberties, originally from 1 (most free) to 7 (least free); scale reversed so that higher values indicate more freedom	1990–2021	Freedom House
Freedom of Academic Exchange	Are scholars free to exchange and communicate research findings, including through media engagement and public lectures? (not = 0, weakly = 1, somewhat = 2, mostly = 3, fully = 4). Official V-Dem variable name: <i>Freedom of Academic Exchange and Dissemination</i>	1990–2021	Coppedge et al. (2023)
IMF Program	Was there an IMF program? Yes = 1	1990–2021	Kentikelenis and Stubbs (2023)
Information Capacity	Latent measure of state information capacity, estimated via item response theory	1990–2012	Brambor et al. (2020)
Log Difference	Number of years elapsed between the vintage and the reference year, logged to capture the nonlinear decay in revision frequency over time	1990–2021	WDI
Media Freedom	Does the government refrain from censoring political information in the media? (not = 0, weakly = 1, somewhat = 2, mostly = 3, fully = 4). Official V-Dem variable name: <i>Government Censorship Effort — Media</i>	1990–2021	Coppedge et al. (2023)
Natural Disaster	Was there a biological (epidemic), climatological (drought, wildfire), meteorological (storm, extreme temperature), hydrological (flood, landslide), or geophysical (earthquake, volcanic activity) disaster? Yes = 1	1990–2021	Centre for Research on the Epidemiology of Disasters (2020)
Opposition Strength	Vote share of the largest opposition party in the most recent legislative election	1990–2021	Cruz et al. (2021)
Polity 2	Revised combined Polity score, from −10 (hereditary monarchy) to +10 (consolidated democracy)	1990–2021	Marshall and Gurr (2020)
Political Corruption Index	Average levels of public sector, executive, legislative, and judicial corruption, from 0 (less corrupt) to 1 (more corrupt)	1990–2021	Coppedge et al. (2023)
Polyarchy	Electoral democracy index, from 0 to 1	1990–2021	Coppedge et al. (2023)

Public Administration	Are public officials rigorous and impartial in the performance of their duties? (no = 0, weakly = 1, modestly = 2, mostly = 3, fully = 4). Official V-Dem variable name: <i>Rigorous and Impartial Public Administration</i>	1990–2021	Coppedge et al. (2023)
Revision	Did the GDP value change between two consecutive WDI releases for the same country-year pair? Yes = 1. Coded NA for the first vintage of each country-year pair	1990–2021	WDI
SNA Change	Was the System of National Accounts in use updated this year? Yes = 1	1994–2021	UN National Accounts Statistics, complemented by WDI Metadata and IMF International Financial Statistics
State Capacity	Latent measure of state capacity combining 21 indicators of extractive, coercive, statistical, informational, and administrative capacity	1990–2015	Hanson and Sigman (2021)
Suffrage	Share of adult citizens that have the legal right to vote in national elections	1990–2021	Coppedge et al. (2023)
Turnover Frequency	Number of changes in party control of the executive in the preceding five years	1990–2021	Cruz et al. (2021)

Figure B.1: Correlation Plot



This figure shows the correlation plot for all variables included in the analysis. To generate this figure, *Abs.% Change (Log)* was coded as zero for all instances where no revision occurred ($Revision = 0$) or where the concept of revision is undefined ($Revision = NA$), in order to include all observations in the correlation matrix. In the actual analysis, *Abs.% Change (Log)* is undefined for non-revisions and the magnitude models are estimated only on observations where a revision occurred and neither GDP value is missing.

Table B.2: Summary Statistics

Variable	N	Mean	Std. Dev.	Min	Pctl. 25	Pctl. 75	Max
Revision, yv	445565						
... 0	372742	83.66%					
... 1	72823	16.34%					
Abs. % Change (Log), yv	72152	-9.358	9.05	-30.02	-17.86	-0.8124	9.769
Polyarchy, y	400255	0.497	0.2742	0.013	0.249	0.762	0.923
Media Freedom, y	385487	2.355	1.409	0	1	3	4
Freedom of Academic Exchange, y	378471	2.858	1.083	0	2	4	4
Suffrage, y	400255	0.9701	0.1615	0	1	1	1
Opposition Strength, y	399810	20.01	22.2	0	0	41.5	99.5
Turnover Frequency, Prev. 5 Yrs, y	399572	0.4904	0.6054	0	0	1	3
Political Corruption Index, y	398683	0.5109	0.3023	0.002	0.207	0.785	0.967
Public Administration, y	400255	2.206	1.094	0	1	3	4
State Capacity, y	369347	0.4132	0.9589	-2.029	-0.2548	0.9487	2.964
Bureaucratic Quality, y	316092	2.17	1.152	0	1	3	4
Information Capacity, y	142520	0.6459	0.1493	0.2387	0.6051	0.7945	1
IMF Program, y	405089						
... 0	271168	66.94%					
... 1	133921	33.06%					
Financial Crisis, y	405089						
... 0	237749	58.69%					
... 1	167340	41.31%					
Natural Disaster, y	405089						
... 0	156987	38.75%					
... 1	248102	61.25%					
Armed Conflict, y	405089						
... 0	298272	73.63%					
... 1	106817	26.37%					
SNA Change, v	412791						
... 0	386234	93.57%					
... 1	26557	6.43%					
Data Management Error, v	529323						
... 0	529272	99.99%					
... 1	51	0.01%					
Log Difference, $v - y$	529323	2.245	0.9377	-1.099	1.792	2.944	3.434

C Additional Descriptive Information

This appendix provides additional context for the main analysis. First, I discuss the threats to validity that create demand for revisions — why countries revise their data. Then, I provide detailed examples of prominent data revisions that illustrate the range of magnitude and direction of revisions in the dataset.

C.1 Why Revise?

Unlike latent concepts like election integrity, state capacity, or democratic consolidation, which rely on expert coding that might be ideologically biased or improperly aggregated ([Giannone, 2010](#); [Martínez i Coma and van Ham, 2015](#); [Hanson and Sigman, 2021](#)), macroeconomic indicators purportedly measure observable, neutral, “objective” outputs. Still, these indicators are often accused of oversimplifying abstract concepts like wealth, inequality, or unemployment ([Mügge, 2022](#)). For example, GDP excludes unpaid household services, which are disproportionately performed by women ([DeRock, 2021](#)). Sometimes different data sources oversimplify the world in different ways. As a result, the two most common data sources in political science and economics — the WDI and the Penn World Table, respectively ([Goes, 2023](#); [Johnson et al., 2013](#)) — provide estimates that can differ by over 25 percent ([Ram and Ural, 2014](#)). Exporters and importers record the same bilateral trade flows differently ([Linsi et al., 2023](#)), and a comparison of export data from two sources — the IMF and the UN Commodity Trade Statistics — concludes that oftentimes the data are not even correlated ([Amin Gutiérrez de Piñeres, 2006](#), 35). Climate aid ([Michaelowa and Michaelowa, 2011](#); [Weikmans and Roberts, 2019](#)) and FDI ([Kerner, 2014](#)) suffer from similar discrepancies. Even within one country, different agencies might compete for data collection, presenting contradictory results ([Pellechio and Cady, 2006](#)). The US is among the few countries without a centralized NSO; its federal statistical system consists of over 100 statistical agencies, units, and programs that might provide conflicting information. And even an oversimplified world is difficult to measure. This section provides more detailed information about the challenges of measuring valid macroeconomic data.

One challenge, as mentioned in the main text, is low statistical capacity. A 2005 survey by the [United Nations Economic Commission for Africa \(2005\)](#) found that some NSOs in the continent had as few as three national accountants. A 2023 survey of 14 NSOs, conducted by the Inter-American Development Bank, found that only half of the employees working with statistical analysis displayed basic competence in probability, descriptive statistics, survey sampling, and arithmetic ([Mejía Guerra et al., 2023](#), 14). Many lacked the expertise to report data consistent with the System of National Accounts (SNA), a global standardization framework developed by the International Comparison Program (ICP) to enable cross-country comparisons. Communist countries did not begin to adopt the SNA until 1993, and North Korea still uses a Marxism-

inspired alternative, the Material Product System (Herrera 2010, 23n8; van Heijster and DeRock 2022, 84n1), rendering its data incomparable to other countries.

Even countries that follow the SNA might struggle with difficult-to-measure concepts like imputed rent, thus underestimating household final consumption expenditure — an important component of GDP (Olinto Ramos et al., 2008). Moreover, it is difficult to quantify the size of the informal economy, which accounts for up to 44 percent of the GDP in the developing world (Coyle, 2014, 110). Population figures tend to be extrapolated from the last census and can grow progressively inaccurate over time; Lebanon, for instance, last conducted a census in 1932 (Devarajan, 2013). Some countries struggle with overly ambitious data-collection efforts: Chile’s 2012 census, which spanned three months, failed to account for nearly 10 percent of the population and was replaced by a single-day census in 2017.

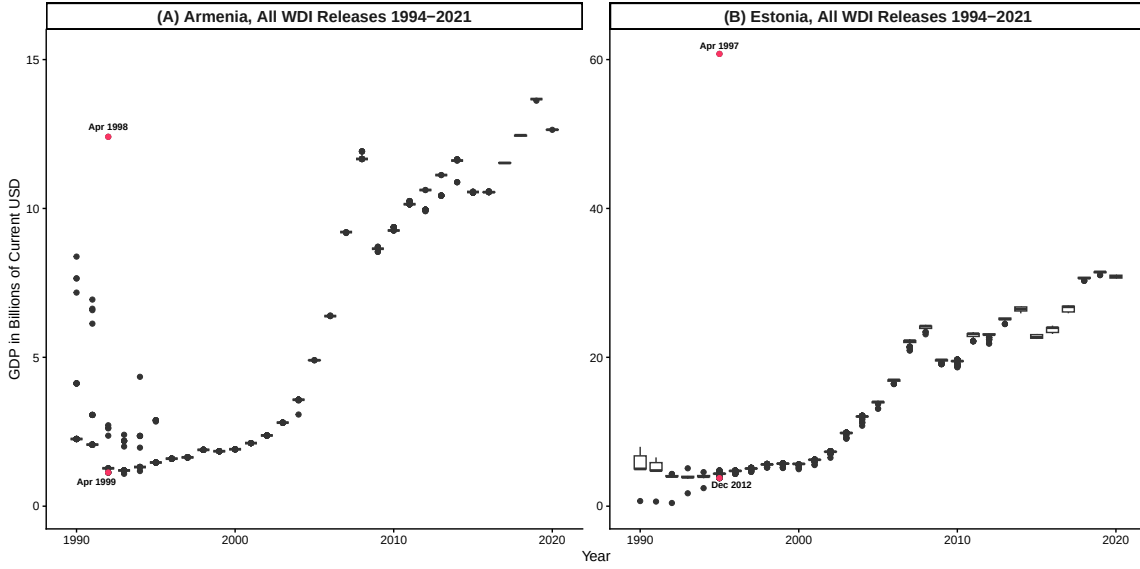
Another challenge to measuring valid macroeconomic data relates to political incentives. In federations like Nigeria, states inflate population figures to receive higher fiscal transfers from the federal government (Devarajan, 2013). Aid-dependent countries systematically understate their finances to appear poor and attract more aid (Kerner et al., 2017). Even industrialized democracies overstate how much climate aid they provide — particularly when domestic constituencies value environmental objectives (Michaelowa and Michaelowa, 2011) — and misrepresent public finance statistics to abide by the rules of the European Union, as Greece did (Alt et al., 2014).

A final challenge relates to idiosyncratic data management errors. These cannot be predicted systematically but are fairly easy to identify. Some trivial errors include: in 2006, “the country names for Burundi and Cameroon are in reverse order;” in 2008, “columns are incorrectly labeled as 1990; data are for 1995;” and in 2010, “an error for Zimbabwe’s data” meant that several indicators, including GDP, “should be presented as not available for all years in the WDI database” (see World Bank 2023 for errata). Below, I provide concrete examples.

Figure C.1 presents the GDP of two former Soviet republics, Armenia and Estonia (both of which gained independence in 1991). According to the April 1998 WDI, Armenia had a GDP of 12.4 billion in 1992 — a number over four times as large as what any other WDI release reports. According to the April 1997 WDI, Estonia had a GDP of 60.8 billion in 1995 — a number at least 13 times as large as what other releases report. Since these extreme values only appear once, I assume they are the product of a data management error corrected in subsequent vintages.

Figure C.2 presents the GDP of two war-plagued countries in the Middle East, Iraq and Syria, and a newly independent country, Timor-Leste. Iraq in 1990 is an interesting case: this country-year pair first enters the WDI in September 1995 and takes the value of 48.66 billion until September 2005, at which point it ceases to be included. It reappears in the December 2014 WDI, at which point it is reported to

Figure C.1: Current GDP of Armenia and Estonia, 1990–2020



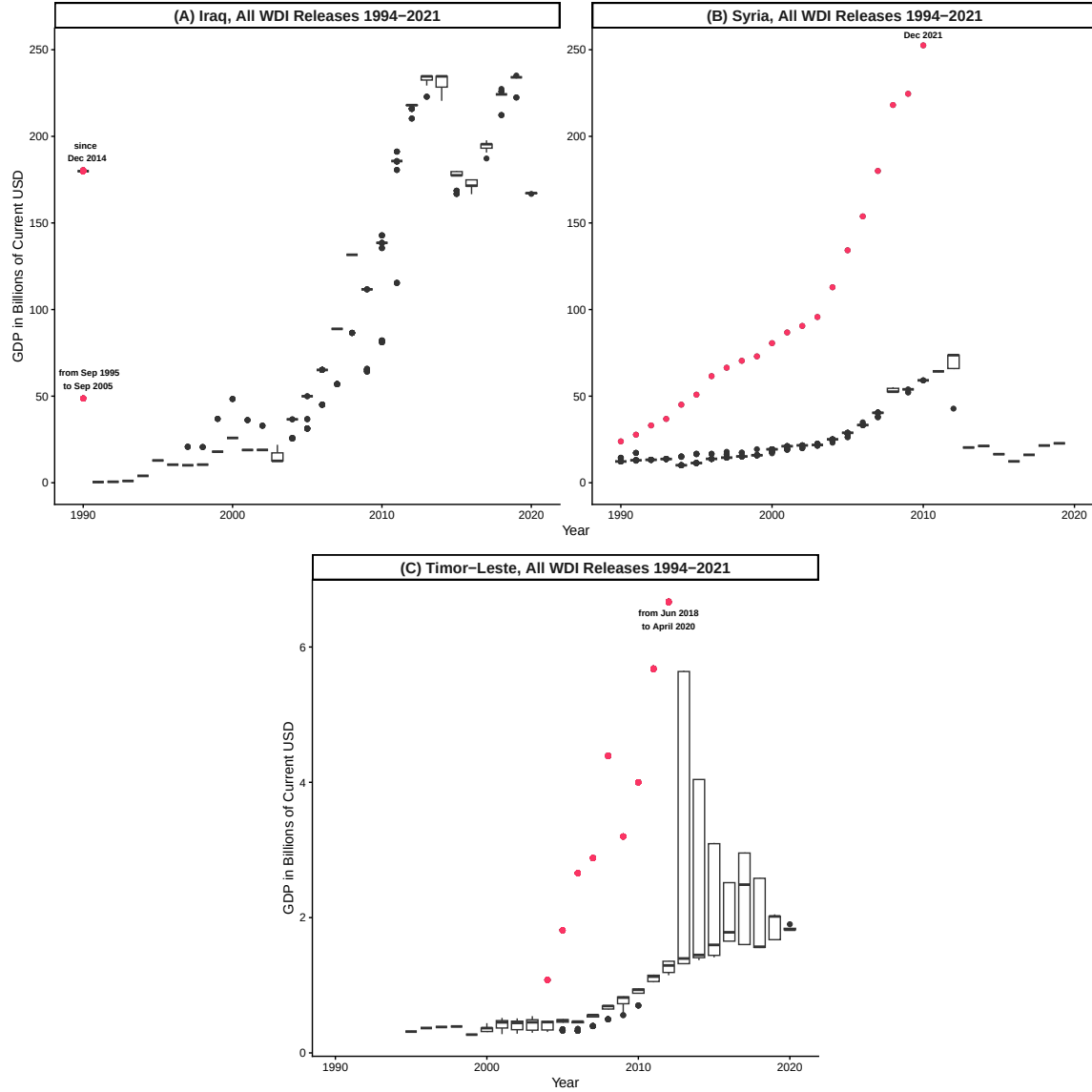
These boxplots present the distribution of current GDP estimates for (A) Armenia and (B) Estonia, from 1990 to 2020, using data drawn from the 104 WDI releases from April 1994 to December 2021.

be nearly four times as large: 179.91 billion. Syria’s GDP from 1990 to 2010 is considerably larger in the December 2021 WDI than in other vintages. As of 2024, these values have not been revised; they are the most up-to-date values. The [World Bank \(2023\)](#) warns that *some* of its data series for Syria “should not be used for analysis or assessment” because they “are incomplete and based on estimates,” but it does not specify *which* data series are affected by these issues.

As Figure C.2 further shows, all vintages report a GDP between 1.1 and 1.2 billion for Timor-Leste in 2012, with the exception of 16 vintages between June 2018 and April 2020 that report a number six times larger. Indeed, these 16 vintages provide exceptionally large values for Timor-Leste for all years from 2004 until 2012.

Iraq, Syria, and Timor-Leste’s unusual values appear in multiple vintages. They are not specifically listed by the World Bank in its Data Updates and Errata website. In April 2022, I raised some of these issues to the WDI team via e-mail; three months later, a member of the World Bank’s Development Data Group responded that they were related to the timing of the IT team’s periodical maintenance and suggested I clear my browser cache, but doing so did not rectify these issues, and the WDI team did not respond to my follow-up inquiries about these specific observations. Thus, I do not code them as an error. In intentionally setting a high bar for an error, I likely underestimate the prevalence of such errors.

Figure C.2: Current GDP of Iraq, Syria, and Timor-Leste, 1990–2020



These boxplots present the distribution of current GDP estimates for (A) Iraq, (B) Syria, and (C) Timor-Leste from 1990 to 2020, using data drawn from the 104 WDI releases from April 1994 to December 2021.

C.2 Prominent Data Revisions

About 39.8 percent of all economies surveyed by the IMF in 2020 first released annual GDP data within 90 days of the reference period, whereas 54.4 percent did so within 91 to 365 days, with no available information for the remaining 5.8 percent (Baer et al., 2022, 17). Many of these economies later revised their preliminary data, as recommended by international organizations. This section provides additional information about prominent data revisions. I begin by providing examples of what Croushore and Stark (2003) deems information-based revisions, which are related to the “discovery” of new data. In these cases, countries revise

their data to incorporate newly available information that allows for more precise measurement.

In 2009, Prime Minister George Papandreou came to power in Greece and requested help from Eurostat and the International Monetary Fund (IMF) to revise the country’s finances.¹ As these revisions quickly established, previous administrations had overestimated tax revenues, disregarded social security and military expenses, and engaged in creative accounting to hide government liabilities. Between April and October 2009, the planned deficit ratio for 2009 was revised from 3.7 to 12.5 percent of the Gross Domestic Product (GDP) (European Commission, 2010), a number that — according to Eurostat — was still far below the truth (Aragão and Linsi, 2022). Following these revisions, credit rating agencies downgraded Greece, which requested multiple IMF and EU loans to avoid default.

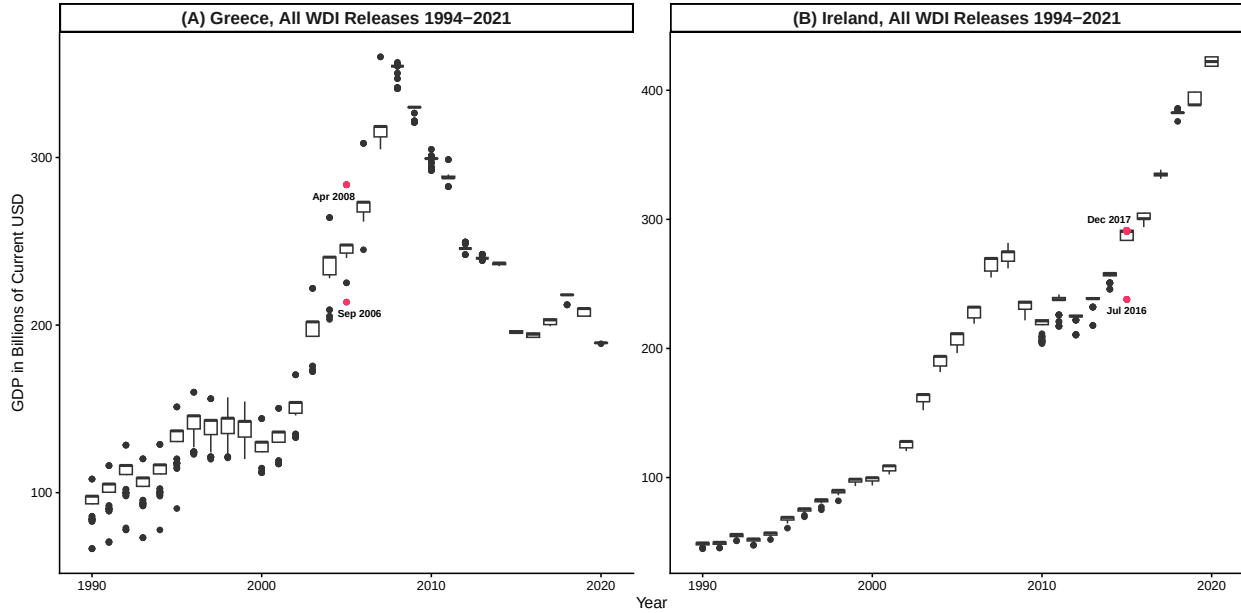
A revision in Ireland caused similar controversy. From 2015 to 2016, the Irish Central Statistics Office reported a GDP growth of 26.3 percent. Citing statistical confidentiality rules, authorities initially refused to release detailed data, instead devising a new statistic — the Modified Gross National Income (GNI*) — to remove “globalization-related” distortions. Ireland’s “leprechaun economics” became a source of ridicule, and the country was blacklisted as a tax haven. Years later, economists discovered that these distortions had been driven by Apple’s decision to onshore intellectual property assets to Ireland in 2015 (Polyak, 2023). Figure C.3 illustrates the cases of Greece and Ireland. Between the September 2006 and April 2008 WDI releases, Greece’s GDP for 2005 increased by nearly 33 percent, from 213.7 billion to 283.7 billion. Between the July 2016 and the December 2017 WDI, Ireland’s 2015 GDP figures increased by 22.6 percent.

Beyond Europe, the Ghana Statistical Service released new GDP estimates in 2010, with support from the Danish International Development Agency and the IMF. After upgrading from the 1968 to the 1993 System of National Accounts (SNA), updating the base year, and disaggregating data by economic sector, it concluded that the country’s GDP was 60.3 percent larger than previously thought (Jerven and Ebo Duncan, 2012). Kenya, Nigeria, and others similarly reported GDP increases after incorporating new information from informal activities (African Development Bank, 2013). In the following years, the World Bank upgraded all three countries from low income to lower middle income economy. This shift was associated with less generous lending terms, as countries with per capita incomes above a certain threshold lose access to concessional lending (Kerner et al., 2017).

Every five to ten years, the International Comparison Program (ICP) surveys how much the same basket of goods costs in different currencies and constructs purchasing power parity (PPP) exchange rates. These exchange rates, in turn, are used to convert SNA data from nominal (current) to PPP terms, which are comparable across borders. Until 1996, ICP price surveys only covered the developed world, making less

¹In fact, information-based revisions are often tied to international actors; for example, subscribing to the IMF’s Special Data Dissemination Standard (SDDS) can increase transparency (Vadlamannati et al., 2018).

Figure C.3: Current GDP of Greece and Ireland, 1990–2020



These boxplots present the distribution of current GDP estimates for (A) Greece and (B) Ireland from 1990 to 2020, using data drawn from the 104 WDI releases from April 1994 to December 2021. The estimate reported for Greece in 2005 is 70 billion dollars (32.78 percent) larger in the April 2008 WDI than in the April 2007 WDI. The estimate reported for Ireland in 2015 is 53.8 billion dollars (22.6 percent) larger in the December 2017 WDI than in the July 2016 WDI. Section 4 discusses the data in detail.

accurate extrapolations for the developing world (Deaton and Aten, 2017). ICP rounds in 2005, 2011, and 2017 reduced uncertainty by incorporating new information from large developing countries, leading to substantial data revisions. However, price surveys in China were only conducted in urban areas, introducing yet another potential measurement issue (Bolt and van Zanden, 2024). Ultimately, ICP rounds still disagree with each other due to differences in relative prices, consumption patterns, region-specific PPP adjustments, and accounting or reporting practices (Deaton and Aten, 2017). While my analysis focuses on current GDP rather than PPP-adjusted figures, ICP rounds illustrate the broader point that structural revisions can be substantial and politically consequential.

Besides information-based revisions, countries may also conduct structural revisions, changing definitions, base years, or aggregation methods. For example, the 1993 SNA introduced the concept of imputed rent, which significantly altered the definition and calculation of GDP (Olinto Ramos et al., 2008). Nicaragua revised its national accounts in 2003, changing the base year from 1980 to 1994 and implementing the 1993 SNA; as a result, the country’s current GDP for the year 2000 increased by 70 percent (Olinto Ramos et al., 2008, 9). In 2014, EU countries revised their GDP definition to include drug trafficking and prostitution; as a result, the Italian and British economies increased by four percent each (Coyle, 2014, 110). For quarterly

data, at least, information-based revisions are most common up to one year after the initial data release. The more time has elapsed since the original data release, the more structural the nature of revisions (Croushore and Stark, 2003).

D Alternative Measures of State Capacity

Table D.1: Determinants of the Likelihood and Magnitude of Data Revisions: Alternative Measures of State Capacity

	Dependent Variable:					
	Revision (1)	Revision (2)	Revision (3)	Abs. % Change (4)	Abs. % Change (5)	Abs. % Change (6)
Polyarchy, y	0.63*** (0.08)	0.69*** (0.09)	0.74*** (0.14)	1.38*** (0.33)	1.56*** (0.36)	2.20*** (0.56)
State Capacity, y	0.03 (0.03)			0.09 (0.13)		
Bureaucratic Quality, y		0.07*** (0.02)			0.11* (0.07)	
Information Capacity, y			0.22 (0.17)			-3.11*** (0.66)
IMF Program, y	0.03 (0.02)	0.02 (0.02)	0.01 (0.03)	-0.13* (0.08)	-0.06 (0.09)	0.07 (0.13)
Financial Crisis, y	-0.01 (0.02)	0.00 (0.02)	-0.08** (0.03)	-0.18** (0.08)	-0.19** (0.09)	-0.16 (0.13)
Natural Disaster, y	0.05*** (0.02)	0.06*** (0.02)	0.03 (0.03)	-0.11 (0.07)	-0.11 (0.08)	-0.09 (0.13)
Armed Conflict, y	0.04 (0.03)	0.03 (0.04)	-0.02 (0.06)	-0.27** (0.14)	-0.22 (0.15)	-0.05 (0.23)
SNA Change, v	0.25*** (0.03)	0.19*** (0.03)	0.42*** (0.05)	1.72*** (0.12)	1.41*** (0.13)	1.70*** (0.18)
Data Management Error, v	6.29*** (1.14)	6.18*** (1.12)	18.37 (665.76)	12.94*** (1.61)	11.07*** (1.53)	15.05*** (1.90)
Log Difference, $v - y$	-1.11*** (0.03)	-1.00*** (0.04)	-1.32*** (0.05)	-4.78*** (0.11)	-4.59*** (0.12)	-3.94*** (0.18)
Intercept	-2.22*** (0.56)	-2.65*** (0.56)	-2.04*** (0.62)	1.05 (0.85)	-0.63 (0.90)	1.68* (0.90)
Log Likelihood	-68628.35	-58222.55	-25724.18	-195065.95	-166407.02	-77091.71
Observations	349320	295337	135206	58505	49945	23213
Number of Countries	162	135	66	162	135	66
Number of Years	27	31	24	27	31	24
Number of Vintages	103	103	103	76	78	71
Variance: Countries	0.24	0.24	0.30	3.25	3.47	4.91
Variance: Years	0.08	0.03	0.16	7.48	11.87	2.15
Variance: Vintages	27.52	27.31	27.78	26.54	22.39	15.25

Notes: This table presents three logistic regressions (Models 1–3) and three linear regressions (Models 4–6) with random intercepts for country, year, and vintage. Models 1 and 4 replace *Public Administration* with the [Hanson and Sigman \(2021\)](#) *State Capacity* index; Models 2 and 5 with the ICRG measure of *Bureaucratic Quality*; Models 3 and 6 with the [Brambor et al. \(2020\)](#) *Information Capacity* index, which covers 85 countries and up to 2012 only. *Data Management Error* produces a large standard error in Model 3 due to near-perfect separation in the restricted sample. y is the reference year; v is the vintage. Standard errors in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

Table D.1 replaces *Public Administration* with three alternative measures of state capacity: the [Hanson and Sigman \(2021\)](#) state capacity index (Models 1 and 4), the ICRG measure of *Bureaucratic Quality* (Models 2 and 5), and the [Brambor et al. \(2020\)](#) information capacity index (Models 3 and 6). The democracy

finding is robust across all six models: *Polyarchy* is positive and highly significant for both likelihood and magnitude regardless of which capacity measure is used, and the coefficients are broadly consistent with the main results. The capacity measures themselves show mixed results: *Bureaucratic Quality* is significant for both outcomes, *State Capacity* is insignificant throughout, and *Information Capacity* is insignificant for likelihood but negative and significant for magnitude. This last result should be interpreted with caution given the substantially restricted sample (66 countries, up to 2012). *Data Management Error* produces a large standard error in Model 3 due to near-perfect separation in this restricted sample.

E Reduced Sample: Only Main Releases

The WDI are typically updated twice a year, with additional updates as needed. As the World Bank shifted from print to digital, the number of annual updates increased: there were ten updates in 2017 (in every month except for January and February), compared to only one each year from 1997 to 2004 (always in April). While the main models include data on all available vintages, Table E.1 uses only each year’s main scheduled release, typically in April.² Restricting the analysis to the main release reduces noise (since the information provided by two consecutive releases is likely very similar) and ensures that recent years (with many releases) do not disproportionately influence the results compared to earlier years (with fewer releases). The democracy finding holds for likelihood across both specifications, but the magnitude results are weaker in this restricted sample, likely reflecting the substantial reduction in observations (70,041 versus 375,052) and vintage variation (26 vintages versus 103). *Data Management Error* is excluded from the likelihood models due to near-perfect separation in the restricted sample, but retained in the magnitude models where it is positive but insignificant, consistent with the small number of documented coding errors among the main annual releases.

²For 2020, I use August instead of April, and for 2021 I use September, due to pandemic-related delays in data collection and dissemination.

Table E.1: Determinants of the Likelihood and Magnitude of Data Revisions: Only Main Releases

	Dependent Variable:			
	Revision (1)	Revision (2)	Abs. % Change (3)	Abs. % Change (4)
Polyarchy, y	0.65*** (0.17)		0.51 (0.49)	
Media Freedom, y		0.06* (0.03)		0.27*** (0.10)
Freedom of Academic Exchange, y		0.16*** (0.05)		−0.42*** (0.15)
Suffrage, y		0.25 (0.25)		0.98 (0.69)
Opposition Strength, y		−0.00*** (0.00)		−0.01*** (0.00)
Turnover Frequency, Prev. 5 Yrs, y		0.02 (0.03)		−0.11 (0.09)
Political Corruption Index, y		−0.38* (0.20)		−2.14*** (0.56)
Public Administration, y	0.07* (0.04)	0.01 (0.05)	0.35*** (0.12)	0.27** (0.14)
IMF Program, y	0.01 (0.04)	0.02 (0.04)	−0.09 (0.12)	−0.06 (0.13)
Financial Crisis, y	0.09** (0.04)	0.08* (0.04)	−0.21 (0.13)	−0.16 (0.14)
Natural Disaster, y	0.01 (0.04)	0.04 (0.04)	−0.12 (0.11)	−0.13 (0.12)
Armed Conflict, y	0.04 (0.07)	0.07 (0.08)	0.05 (0.20)	0.15 (0.22)
SNA Change, v	−0.15** (0.07)	−0.19** (0.08)	0.85*** (0.23)	0.98*** (0.24)
Data Management Error, v			6.05 (4.33)	5.50 (4.33)
Log Difference, $v - y$	−1.45*** (0.07)	−1.38*** (0.07)	−4.18*** (0.20)	−4.09*** (0.21)
Intercept	1.43* (0.81)	1.21 (0.88)	−0.75 (0.99)	0.53 (1.22)
Log Likelihood	−16986.21	−15607.55	−65198.06	−59211.33
Observations	70041	64459	20114	18273
Number of Countries	169	166	168	164
Number of Years	31	30	30	29
Number of Vintages	26	26	25	25
Variance: Countries	0.58	0.61	3.03	3.09
Variance: Years	0.72	0.56	5.98	5.42
Variance: Vintages	15.91	17.05	15.18	14.60

Notes: This table presents two logistic regressions (Models 1–2) and two linear regressions (Models 3–4) with random intercepts for country, year, and vintage, estimated on the subset of vintages corresponding to the main scheduled annual release (typically April; August for 2020 and September for 2021 due to pandemic-related delays). *Data Management Error* is excluded from the likelihood models (Models 1–2) due to near-perfect separation in the restricted sample — the same issue that arises in Model 2 of the main analysis — but is retained in the magnitude models (Models 3–4), consistent with the main models. Its coefficient is positive but insignificant in both magnitude models, likely reflecting the small number of documented coding errors in the April-only sample. y is the reference year; v is the vintage. Standard errors in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

F Models With Fixed Effects

Table F.1 replaces the random effects specification with country, year, and vintage fixed effects. Given the large number of units (169 countries), time periods (31 years), and vintages (103), these models are computationally inefficient: they lead to a substantial loss of degrees of freedom, and *Log Difference* drops

Table F.1: Determinants of the Likelihood and Magnitude of Data Revisions: Fixed Effects

	Dependent Variable:			
	Revision (1)	Revision (2)	Abs. % Change (3)	Abs. % Change (4)
Polyarchy, y	0.53** (0.24)		0.71 (0.72)	
Media Freedom, y		0.05 (0.03)		0.33*** (0.12)
Freedom of Academic Exchange, y		0.09 (0.06)		-0.26 (0.24)
Suffrage, y		0.08 (0.36)		0.75 (1.03)
Opposition Strength, y		-0.00 (0.00)		-0.01** (0.00)
Turnover Frequency, Prev. 5 Yrs, y		0.00 (0.02)		0.07 (0.10)
Political Corruption Index, y		0.35 (0.25)		0.16 (0.91)
Public Administration, y	0.02 (0.05)	0.02 (0.05)	-0.01 (0.12)	-0.01 (0.13)
IMF Program, y	0.02 (0.03)	0.02 (0.03)	-0.03 (0.13)	-0.02 (0.14)
Financial Crisis, y	-0.01 (0.04)	-0.02 (0.04)	-0.13 (0.14)	-0.17 (0.14)
Natural Disaster, y	0.05** (0.02)	0.06*** (0.02)	-0.14* (0.07)	-0.14* (0.07)
Armed Conflict, y	0.02 (0.07)	0.01 (0.07)	-0.31 (0.27)	-0.23 (0.31)
SNA Change, v	0.22** (0.10)	0.24** (0.10)	1.65*** (0.40)	1.78*** (0.40)
Data Management Error, v	6.42*** (0.58)	13.59*** (0.16)	11.74*** (1.00)	11.32*** (1.01)
Observations	267691	248755	62973	58655
Number of Countries	169	166	169	166
Number of Years	31	30	31	30
Number of Vintages	78	77	78	78
Log Likelihood	-74334.04	-69562.96		
Pseudo R^2	0.49	0.48		
R^2 (full model)			0.43	0.43
R^2 (proj. model)			0.00	0.01

Notes: This table presents two logistic regressions (Models 1–2) and two linear regressions (Models 3–4) with fixed effects for country, year, and vintage. Standard errors clustered by country. *Log Difference* is excluded from all models due to collinearity with vintage fixed effects. The likelihood models drop observations from groups with no within-group variation in the outcome (incidental parameters problem), reducing the sample from 375,052 to 267,691 observations. y is the reference year; v is the vintage. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

out due to collinearity with the vintage fixed effects. The models are not directly comparable to the random effects models because of the reduced sample size. This problem is especially pronounced for the likelihood models (Models 1 and 2), which suffer from the incidental parameters problem: observations from groups with no within-group variation in the outcome are automatically dropped. Fixed effects models also discard between-country variation and focus exclusively on within-country changes over time, which is particularly costly for slowly-changing variables like democracy. Despite these limitations, the main democracy finding survives for likelihood: *Polyarchy* is positive and significant in Model 1 ($p < 0.05$). The magnitude result

is less robust, which is expected given that the between-country variation in democracy is absorbed by the country fixed effects. Among the mechanism indicators, *Media Freedom* is significant for magnitude ($p < 0.01$) but not for likelihood, while *Freedom of Academic Exchange* is insignificant throughout, consistent with the limited within-country variation in these indicators over the sample period.

G Direction of Revisions

Table G.1: Direction of Revisions

	Dependent Variable:	
	Upward Revision	Upward Revision
	(1)	(2)
Polyarchy, y	0.16** (0.07)	
Media Freedom, y		0.01 (0.02)
Freedom of Academic Exchange, y		0.01 (0.02)
Suffrage, y		-0.09 (0.09)
Opposition Strength, y		0.00 (0.00)
Turnover Frequency, Prev. 5 Yrs, y		-0.01 (0.02)
Political Corruption Index, y		-0.12 (0.08)
Public Administration, y	0.01 (0.02)	0.00 (0.02)
IMF Program, y	0.00 (0.02)	0.01 (0.02)
Financial Crisis, y	-0.05** (0.02)	-0.03 (0.02)
Natural Disaster, y	0.00 (0.02)	0.01 (0.02)
Armed Conflict, y	-0.01 (0.03)	-0.02 (0.03)
SNA Change, v	0.15*** (0.04)	0.15*** (0.04)
Data Management Error, v	0.77 (0.49)	0.93* (0.52)
Log Difference, $v - y$	-0.03*** (0.01)	-0.04*** (0.01)
Intercept	0.24*** (0.07)	0.43*** (0.13)
Log Likelihood	-42079.88	-39170.72
Observations	62973	58655
Number of Countries	169	166
Number of Years	31	30
Number of Vintages	78	78
Variance: Countries	0.02	0.02
Variance: Years	0.00	0.00
Variance: Vintages	0.23	0.22

Notes: Logistic regressions with random intercepts for country, year, and vintage, estimated on observations where a revision occurred and neither GDP value is missing (*Abs. % Change* > 0). The outcome equals 1 if the revision is upward and 0 if downward. Appearances and disappearances are excluded since the direction of revision is undefined when one value is missing. y is the reference year; v is the vintage. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

Table G.1 examines the direction of revisions, asking whether democracies are more likely to revise their GDP estimates upward rather than downward. The analysis is restricted to observations where a revision occurred and neither GDP value is missing, excluding appearances and disappearances where direction is undefined. Model 1 shows that higher Polyarchy scores are associated with a significantly higher probability of upward revisions ($p < 0.05$), consistent with the expectation that democratic governments initially publish cautious, understated estimates that are subsequently corrected upward as better data becomes available. This result is also consistent with Martínez’s (2022) finding that autocrats tend to inflate their numbers from the outset.

Model 2 decomposes this finding across the democracy mechanism indicators, but none reach conventional significance levels, suggesting that the upward revision tendency is a property of democratic governance broadly rather than any specific institutional channel. SNA changes are consistently associated with upward revisions across both models, reflecting that methodological updates to national accounts (such as the adoption of new SNA standards or the inclusion of previously unmeasured economic activities) tend to increase GDP estimates rather than reduce them. Relatedly, the negative and significant coefficient on *Log Difference* suggests that upward revisions are disproportionately concentrated in the period shortly after initial data release, when new survey information and administrative data are most likely to prompt upward corrections to initially cautious estimates. As more time passes, the remaining revisions become less systematically directional

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